

Práctica 2

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1. Sean X, Y dos variables aleatorias continuas con función de densidad conjunta

$$f_{X,Y}(x,y) = \begin{cases} k & \text{si } 0 < y < x < 1 \\ 0 & \text{En otro caso} \end{cases}$$

- Determine el valor de k .
- Determine $f_X(x)$ y $f_Y(y)$.
- ¿Son X e Y variables aleatorias independientes?
- Determine $F_X(x)$ y $F_Y(y)$.
- Determine $f_{X|Y}(x|y)$ y $f_{Y|X}(y|x)$.
- Determine $F_{X|Y}(x|y)$ y $F_{Y|X}(y|x)$.
- Determine $P(Y \leq 0.1 | X = 0.4)$.
- Determine $P(X \leq 0.4 | Y \leq 0.1)$.

2. Sean X, Y dos variables aleatorias continuas con función de densidad conjunta

$$f_{X,Y}(x,y) = \begin{cases} kxy & \text{si } 0 < y < x < 1 \\ 0 & \text{En otro caso} \end{cases}$$

- Determine el valor de k .
- Determine $f_X(x)$ y $f_Y(y)$.
- ¿Son X e Y variables aleatorias independientes?
- Determine $P(X + Y > 1)$.
- Determine $P(Y \leq X^2)$.

3. Sean X, Y dos variables aleatorias continuas con función de densidad conjunta

$$f_{X,Y}(x,y) = \begin{cases} x+y & \text{si } 0 < x < 1, 0 < y < 1 \\ 0 & \text{En otro caso} \end{cases}$$

- Determine $f_X(x)$ y $f_Y(y)$.
- ¿Son X e Y variables aleatorias independientes?
- Determine $F_X(x)$ y $F_Y(y)$.
- Determine $f_{X|Y}(x|y)$ y $f_{Y|X}(y|x)$.
- Determine $F_{X|Y}(x|y)$ y $F_{Y|X}(y|x)$.
- Determine $F_{X,Y}(x,y)$.

4. La cantidad en miligramos de dos componentes de un producto es una variable aleatoria bidimensional cuya función de densidad conjunta viene dada por

$$f_{X,Y}(x,y) = \begin{cases} 4xy & \text{si } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{En otro caso} \end{cases}$$

- Determine $f_X(x)$ y $f_Y(y)$.
- ¿Son X e Y variables aleatorias independientes?
- Determine $F_X(x)$ y $F_Y(y)$.
- Determine $P(X + Y \leq 0.5)$.
- Determine $P(X \leq 0.3 | Y = 0.8)$.

5. Considere las variables aleatorias X, Y con función de densidad conjunta dada por

$$f_{X,Y}(x,y) = \begin{cases} 6 & \text{si } 0 \leq x \leq 1, x^2 \leq y \leq x \\ 0 & \text{En otro caso} \end{cases}$$

- Determine $f_X(x)$ y $f_Y(y)$.
- ¿Son X e Y variables aleatorias independientes?

6. Considere las variables aleatorias X, Y con función de densidad conjunta dada por

si sale

$$f_{X,Y}(x,y) = \begin{cases} 30xy^2 & \text{si } 0 \leq x \leq 1, x-1 \leq y \leq 1-x \\ 0 & \text{En otro caso} \end{cases}$$

no sale

- Determine $f_X(x)$ y $f_Y(y)$.
- Determine $P(X > Y)$.
- Determine $F_{Y|X}(y|x)$.
- Determine $P(Y > 0 | X = \frac{3}{4})$.
- ¿Son X e Y variables aleatorias independientes?

7. Considere las variables aleatorias X, Y con función de densidad conjunta dada por

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{y} e^{-\frac{x}{y}-y} & \text{si } x > 0, y > 0 \\ 0 & \text{En otro caso} \end{cases}$$

no sale

- Determine $f_X(x)$ y $f_Y(y)$.
- Determine $P(X < 2Y)$.
- Determine $P(Y < 2X)$.

8. Considere las variables aleatorias X, Y con función de distribución conjunta dada por

$$F_{X,Y}(x,y) = \begin{cases} 0 & \text{si } x \leq 0, y \leq 0 \\ \frac{1}{2}(x^3y + xy^3) & \text{si } 0 < x < 1, 0 < y < 1 \\ \frac{1}{2}(x^3 + x) & \text{si } 0 < x < 1, y > 1 \\ \frac{1}{2}(y + y^3) & \text{si } 0 < y < 1, x > 1 \\ 1 & \text{si } x \geq 1, y \geq 1 \end{cases}$$

- Determine $f_{X,Y}(x,y)$.
- Determine $P(2X < 3Y)$.

9. Sean X, Y dos variables aleatorias continuas independientes e igualmente distribuidas como exponenciales de parámetros λ_1, λ_2 , respectivamente.

- Determine $P(X < Y)$.

10. Sean X, Y dos variables aleatorias continuas independientes e igualmente distribuidas como exponenciales de parámetro λ .

- Determine $P(X < 2Y)$.
- Determine $P(X + Y \leq 1, X - Y \leq 1)$.

11. Sea X la cantidad de tiempo que una persona espera para ser atendida en una fila. Cuanto más rápido trabaje el servidor en la fila, menor será el tiempo de espera. Sea que Y la velocidad a la que trabaja el servidor, que será desconocida. La función de densidad condicional de X dado $Y = y$ es

$$f_{X|Y}(x|y) = \begin{cases} ye^{-xy} & \text{si } x \geq 0 \\ 0 & \text{en otro caso} \end{cases}.$$

Si la densidad marginal de Y es $f_Y(y) = e^{-y}$ para $y > 0$. Determine $f_{X,Y}(x, y)$.

12. Un ordenador tarda un total de Y segundos en procesar un mensaje de correo electrónico, esta incluye el tiempo X durante el cual el emnsaje está en cola esperando ser procesado ($Y \geq X$). La función de densidad conjunta es

$$f_{X,Y}(x, y) = e^{-y}, \quad 0 \leq x \leq y < \infty$$

Determine la probabilidad de que un mensaje haya estado a lo sumo un segundo en la cola si el tiempo total que ha durado su procesamiento ha sido al menos dos segundos, es decir, $P(X \leq 1 | Y \geq 2)$.

13. Si X es el tiempo que transcurre desde el momento en que un cliente llega a una tienda hasta que abandona la ventanilla de servicio y Y es el tiempo que el cliente emplea haciendo fila para ser atendido, la función de densidad conjunta de estas variables aleatorias viene dada por:

$$f_{X,Y}(x, y) = \begin{cases} e^{-x} & \text{si } 0 \leq y \leq x \\ 0 & \text{en otro caso} \end{cases}.$$

Defina la variables $W = X - Y$.

- Determine $E(W)$ y $Var(W)$.
- Determine $P(W \geq 4)$.

14. Considere las variables aleatorias X, Y con función de densidad conjunta dada por

$$f_{X,Y}(x, y) = \begin{cases} e^{-x-y} & \text{si } x > 0, y > 0 \\ 0 & \text{En otro caso} \end{cases}.$$

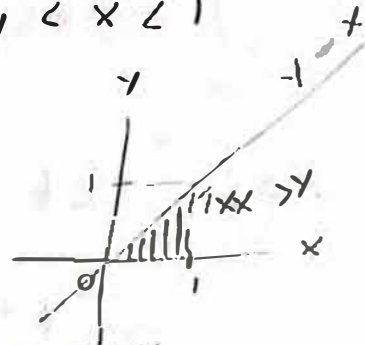
- Determine $f_X(x)$ y $f_Y(y)$.
- Determine $P(0 < X < 1, 1 \leq Y < 2)$.
- Determine $F_{Y|X}(y|x = 3)$
- Determine $E(X^2 Y)$.
- Determine $Cov(X, Y)$.
- ¿Son X e Y incorrelacionadas?

15. Sean X, Y dos variables aleatorias continuas con función de densidad conjunta

$$f_{X,Y}(x,y) = \begin{cases} \frac{9}{2} x^2 y^2 & \text{si } -1 \leq y \leq x \leq 1 \\ 0 & \text{en otro caso} \end{cases}.$$

- a) Determine $f_{Y|X}(y|x)$.
- b) Determine $P(Y \leq 0.1 | X \geq 0.6)$.
- c) Determine $P(X + Y \leq 0)$.
- d) Determine $\text{Cov}(X, Y)$.
- e) Determine $\rho(X, Y)$.

$$1. f_{X,Y}(x,y) = \begin{cases} K & \text{s. } 0 < y < x < 1 \\ 0 & \text{en o.c.} \end{cases}$$



$$a) \int_0^1 \int_0^x K \, dy \, dx = 1$$

$$\Rightarrow \int_0^1 K \cdot y \Big|_0^x \, dx = 1$$

$$\Rightarrow \int_0^1 Kx \, dx = 1$$

$$\Rightarrow K \cdot \frac{x^2}{2} \Big|_0^1 = 1$$

$$\Rightarrow \frac{K}{2} = 1$$

$$\boxed{\Rightarrow K = 2,,}$$

$$\int_0^1 \int_0^x K \, dy \, dx$$

$$\int_0^1 \underbrace{\int_y^1 K \, dx}_{f_Y(y)} \, dy$$

c) X, Y son indep. $\Leftrightarrow$ se cumple:

$$f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$$

$$2 \neq 2x(2-2y)$$

$\therefore X, Y$ no son indep.

$$b) f_X(x) = \int_0^x 2 \, dy$$

$$= 2 \cdot y \Big|_0^x$$

$$\boxed{= 2x,,}$$

$$f_Y(y) = \int_y^1 K \, dx$$

$$= K \cdot x \Big|_y^1$$

$$= K - Ky$$

$$\boxed{= 2 - 2y,,}$$

Práctica 2

Parte 1

$$\begin{aligned}
 d) F_X(x) &= \int_{-\infty}^x f_X(x) dt & F_Y(y) &= \int_{-\infty}^y f_Y(y) dt \\
 &= \int_0^x 2t dt & &= \int_0^y (2 - 2t) dt \\
 &= 2 \cdot \frac{t^2}{2} \Big|_0^x & &= 2 \cdot t - 2 \cdot \frac{t^2}{2} \Big|_0^y \\
 &= \boxed{x^2} & &= \boxed{2y - y^2}
 \end{aligned}$$

$$\begin{aligned}
 e) f_{X|Y}(x|y) &= \frac{f_{X,Y}(x,y)}{f_Y(y)} & f_{Y|X}(y|x) &= \frac{f_{X,Y}(x,y)}{f_X(x)} \\
 &= \frac{2}{2-2y} & &= \frac{2}{2x} \\
 &= \frac{2}{2(1-y)} & &= \boxed{\frac{1}{x}} \\
 &= \boxed{\frac{1}{1-y}}
 \end{aligned}$$

$$\begin{aligned}
 f) F_{X|Y}(x|y) &= \int_{-\infty}^x \frac{f_{X,Y}(x,y)}{f_Y(y)} dt \\
 &= \int_0^x \frac{1}{1-y} dt \\
 &= \frac{1}{1-y} \cdot t \Big|_0^x \\
 &= \boxed{\frac{x}{1-y}}
 \end{aligned}$$

$$\begin{aligned}
 F_{Y|X}(y|x) &= \int_{-\infty}^y \frac{f_{X,Y}(x,y)}{f_X(x)} dt \\
 &= \int_0^y \frac{1}{x} dt \\
 &= \frac{1}{x} \cdot t \Big|_0^y \\
 &= \boxed{\frac{y}{x}}
 \end{aligned}$$

$$g) P(Y \leq 0.1 | X = 0.1)$$

$$= \int_0^{0.1} z \, dy$$

2 · 0.1

$$= \frac{\frac{1}{5}}{\frac{2}{5}}$$

$$= \frac{1}{2}$$

$$h) P(X \leq 0.1 | Y \leq 0.1)$$

$$= \int_0^{0.1} \int_0^{0.1} z \, dx \, dy$$

$\int_0^{0.1} (z - zy) \, dy$

$$= \frac{\frac{2}{25}}{\frac{19}{100}}$$

$$= \frac{8}{19} \approx 0.4211$$

$$2. f_{X,Y}(x,y) = \begin{cases} kxy & \text{si } 0 < y < x < 1 \\ 0 & \text{en o.c.} \end{cases}$$

$$a) \int_0^1 \int_0^x kxy \, dy \, dx = 1$$

$$\Rightarrow \int_0^1 kx \cdot \frac{y^2}{2} \Big|_0^x \, dx = 1$$

$$\Rightarrow \int_0^1 kx \cdot \frac{x^2}{2} \, dx = 1$$

$$\Rightarrow \frac{k}{2} \cdot \frac{x^3}{3} \Big|_0^1 = 1$$

$$\Rightarrow \frac{k}{8} = 1$$

$$\boxed{\Rightarrow k = 8,,}$$

$$b) f_X(x) = \int_0^x 8xy \, dy$$

$$= 8x \cdot \frac{y^2}{2} \Big|_0^x$$

$$= 4x \cdot \frac{x^2}{1} = 4x^3$$

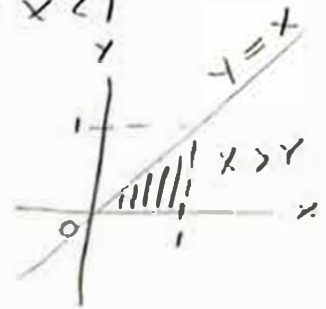
$$\boxed{= 4x^3,,}$$

$$f_Y(y) = \int_y^1 8xy \, dx$$

$$= 8y \cdot \frac{x^2}{2} \Big|_y^1$$

$$= 4y(1 - y^2)$$

$$\boxed{= 4y - 4y^3,,}$$



$$\int_0^1 \int_0^x kxy \, dy \, dx$$

$$\underbrace{\hspace{10em}}_{f_X(x)}$$

$$\int_0^1 \int_y^1 kxy \, dx \, dy$$

$$\underbrace{\hspace{10em}}_{f_Y(y)}$$

c) X, Y son indep. $\Leftrightarrow$ se cumple
que $f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$

$$8xy \neq 4x^3 \cdot 4y - 4y^3$$

$\therefore X, Y$ no son indep.

$$d. P(X + Y > 1)$$

$$= P(Y > 1 - X)$$

$$= \int_{\frac{1}{2}}^1 \int_{1-x}^x xy \, dy \, dx.$$

$$= \int_{\frac{1}{2}}^1 8x \cdot \frac{y^2}{2} \Big|_{1-x}^x \, dx.$$

$$= \int_{\frac{1}{2}}^1 4x \cdot [x^2 - (1-x)^2] \, dx.$$

$$= \int_{\frac{1}{2}}^1 4x [x^2 - (1 - 2x + x^2)] \, dx.$$

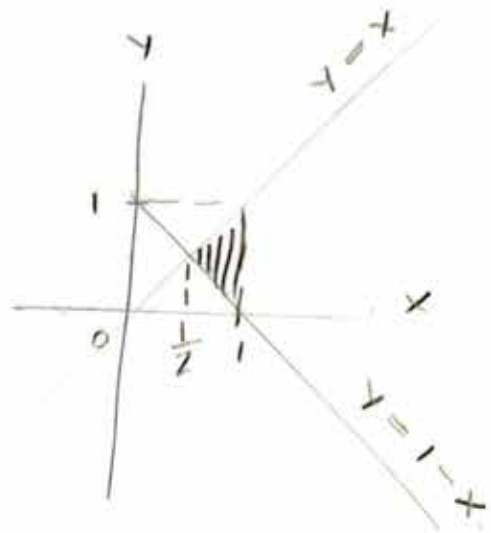
$$= \int_{\frac{1}{2}}^1 4x [x^2 - 1 + 2x - x^2] \, dx$$

$$= \int_{\frac{1}{2}}^1 -4x + 8x^2 \, dx.$$

$$= -4 \cdot \frac{x^2}{2} + 8 \cdot \frac{x^3}{3} \Big|_{\frac{1}{2}}^1$$

$$= \frac{2}{3} - - \frac{1}{6}$$

$$= \boxed{\frac{5}{6}}$$



$$e) P(Y \leq X^2)$$

$$= \int_0^1 \int_0^{x^2} 8xy \, dy \, dx$$

$$= \int_0^1 8x \cdot \frac{y^2}{2} \Big|_0^{x^2} dx$$

$$= \int_0^1 4x \cdot x^2 dx$$

$$= 4 \cdot \frac{x^3}{3} \Big|_0^1$$

$$= \frac{4}{3}$$

$$\boxed{= \frac{2}{3}}$$



$$3. f_{X,Y}(x,y) = \begin{cases} x+y & \text{si } 0 < x < 1, 0 < y < 1 \\ 0 & \text{en o. c.} \end{cases}$$

$$a) f_X(x) = \int_0^1 x+y \, dy$$

$$= x \cdot y + \frac{y^2}{2} \Big|_0^1$$

$$= \frac{x}{1} + \frac{1}{2}$$

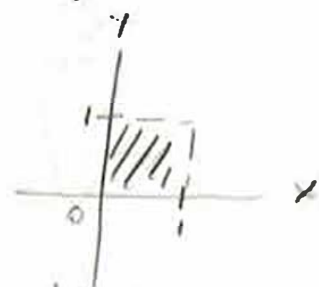
$$\left| = \frac{2x+1}{2} \right|$$

$$f_Y(y) = \int_0^1 x+y \, dx$$

$$= \frac{x^2}{2} + y \cdot x \Big|_0^1$$

$$= \frac{1}{2} + \frac{y}{1}$$

$$\left| = \frac{1+2y}{2} \right|$$



$$\int_0^1 \int_0^1 x+y \, dy \, dx$$

$f_X(x)$

$$\int_0^1 \int_0^1 x+y \, dx \, dy$$

$f_Y(y)$

$$b) X, Y \text{ son indep.} \Leftrightarrow f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$$

$$x+y \neq \left(\frac{2x+1}{2} \right) \left(\frac{1+2y}{2} \right)$$

$\therefore X, Y$ no son indep.

$$\begin{aligned}
 c) F_x(x) &= \int_{-\infty}^x f_x(x) dt \\
 &= \int_0^x 2 \frac{t+1}{2} dt \\
 &= \frac{1}{2} \cdot x \cdot \frac{t^2}{2} + t \Big|_0^x \\
 &= \frac{1}{2} x^2 + x.
 \end{aligned}$$

$$\boxed{= \frac{x^2 + x}{2}}$$

$$\begin{aligned}
 F_y(y) &= \int_{-\infty}^y f_y(y) dt \\
 &= \int_0^y 1 + 2 \frac{t}{2} dt \\
 &= \frac{1}{2} \cdot t + 2 \cdot \frac{t^2}{2} \Big|_0^y \\
 &= \frac{1}{2} \cdot y + y^2
 \end{aligned}$$

$$\boxed{= \frac{y + y^2}{2}}$$

$$d) f_{x|y}(x|y) = \frac{f_{x,y}(x,y)}{f_y(y)}$$

$$= \frac{x+y}{1+2y}$$

$$= \left| \frac{\frac{z}{2x+2y}}{1+2y} \right|$$

$$f_{y|x}(y|x) = \frac{f_{x,y}(x,y)}{f_x(x)}$$

$$= \frac{x+y}{2x+1}$$

$$= \left| \frac{\frac{z}{2x+1}}{2x+1} \right|$$

$$e) F_{x|y}(x|y) = \int_{-\infty}^x \frac{f_{x,y}(x,y)}{f_y(y)} dt$$

$$= \int_0^x \frac{z \cdot t + 2y}{1+2y} dt$$

$$= \frac{1}{1+2y} \left[z \cdot \frac{t^2}{2} + 2y \cdot t \right]_0^x$$

$$= \frac{1}{1+2y} (x^2 + 2yx)$$

$$= \left| \frac{x^2 + 2xy}{1+2y} \right|$$

$$F_{Y|X}(Y|X) = \int_{-\infty}^Y \frac{f_{X,Y}(x,y)}{f_X(x)} dt$$

$$= \int_0^Y \frac{2x+2t}{2x+1} dt$$

$$= \frac{1}{2x+1} \cdot \left(2x \cdot t + 2 \cdot \frac{t^2}{2} \right) \Big|_0^Y$$

$$= \frac{1}{2x+1} \cdot (2xy + y^2)$$

$$= \boxed{\frac{2xy + y^2}{2x+1}}$$

$$f) F_{x,y}(x,y) = \begin{cases} 0 & \text{si } x < 0, y < 0 \\ \frac{x^2 y + y^2 x}{2} & \text{si } 0 \leq x < 1, 0 \leq y < 1 \\ \frac{x^2 + x}{2} & \text{si } 0 \leq x < 1, y \geq 1 \\ \frac{y + y^2}{2} & \text{si } 0 \leq y < 1, x \geq 1 \\ 1 & \text{si } x \geq 1, y \geq 1 \end{cases}$$

$$\begin{aligned} & \int_0^y \int_0^x t + s \, dt \, ds \\ &= \int_0^y \left. \frac{t^2}{2} + s \cdot t \right|_0^x ds \\ &= \int_0^y \frac{x^2}{2} + s x \, ds \\ &= \left. \frac{x^2}{2} \cdot s + \frac{s^2}{2} \cdot x \right|_0^y \\ &= \frac{x^2 y}{2} + \frac{y^2 x}{2} \\ &= \boxed{\frac{x^2 y + y^2 x}{2}} \end{aligned}$$

$$4. f_{X,Y}(x,y) = \begin{cases} 4xy & \text{si } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{en o.c.} \end{cases}$$

$$a) f_X(x) = \int_0^1 4xy \, dy$$

$$= 4x \cdot \frac{y^2}{2} \Big|_0^1$$

$$= \frac{2x}{1}$$

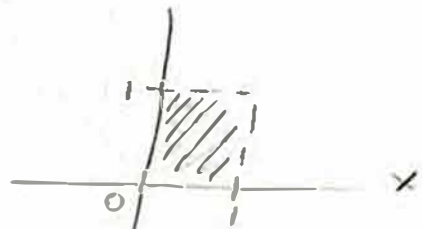
$$\boxed{= 2x,}$$

$$f_Y(y) = \int_0^1 4xy \, dx$$

$$= 4y \cdot \frac{x^2}{2} \Big|_0^1$$

$$= \frac{2y}{1}$$

$$\boxed{= 2y,}$$



$$\int_0^1 \int_0^1 4xy \, dy \, dx$$

$f_X(x)$

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

$f_Y(y)$

$$b) X, Y \text{ son indep.} \Leftrightarrow f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$$

$$4xy = 2x \cdot 2y$$

$$4xy = 4xy$$

$\therefore X, Y$ si son independientes

$$c) F_X(x) = \int_{-\infty}^x f_X(x) \, dt$$

$$= \int_0^x 2t \, dt$$

$$= 2 \cdot \frac{t^2}{2} \Big|_0^x$$

$$\boxed{= x^2,}$$

$$F_Y(y) = \int_{-\infty}^y f_Y(y) \, dt$$

$$= \int_0^y 2t \, dt$$

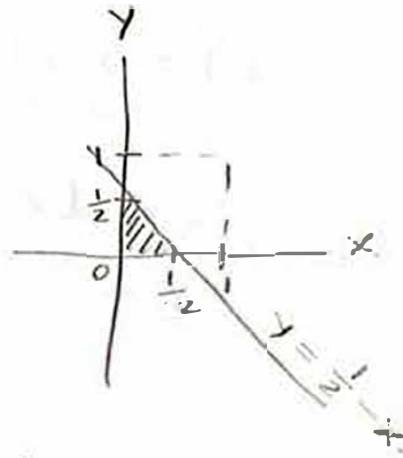
$$= 2 \cdot \frac{t^2}{2} \Big|_0^y$$

$$\boxed{= y^2,}$$

$$d) P(X+Y \leq 0.5)$$

$$= P(Y \leq 0.5 - X)$$

$$= \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}-x} 4xy \, dy \, dx.$$



$$= \int_0^{\frac{1}{2}} 4x \cdot \frac{y^2}{2} \Big|_0^{\frac{1}{2}-x} dx$$

$$= \int_0^{\frac{1}{2}} 2x \left(\frac{1}{2} - x \right)^2 dx.$$

$$= \int_0^{\frac{1}{2}} 2x \left(\frac{1}{4} - 2 \cdot \frac{1}{2} \cdot x + x^2 \right) dx$$

$$= \int_0^{\frac{1}{2}} 2x \left(\frac{1}{4} - x + x^2 \right) dx$$

$$= \int_0^{\frac{1}{2}} \left(\frac{2x}{4} - 2x^2 + 2x^3 \right) dx$$

$$= \frac{1}{2} \cdot \left(\frac{x^2}{2} - 2 \cdot \frac{x^3}{3} + 2 \cdot \frac{x^4}{4} \right) \Big|_0^{\frac{1}{2}}.$$

$$\boxed{= \frac{1}{96}}$$

$$e) P(X \leq 0.3 | Y = 0.8)$$

$$= \frac{\int_0^{0.3} 1 \cdot x \cdot 0.8 \, dx}{2 \cdot 0.8}$$

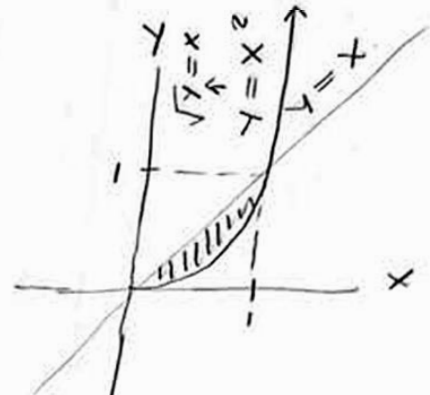
$$= \frac{18}{125} \cdot \frac{8}{5}$$

$$= 0.096$$

$$5. f_{X,Y}(x,y) = \begin{cases} 6 & \text{si } 0 \leq x \leq 1, x^2 \leq y \leq x \\ 0 & \text{en o. c.} \end{cases}$$

$$\begin{aligned} a) f_X(x) &= \int_{x^2}^x 6 \, dy \\ &= 6 \cdot y \Big|_{x^2}^x \\ &= 6x - 6x^2 \\ &= \underline{6x(1-x)} \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \int_y^{\sqrt{y}} 6 \, dx \\ &= 6 \cdot x \Big|_y^{\sqrt{y}} \\ &= 6\sqrt{y} - 6y \\ &= \underline{6(\sqrt{y} - y)} \end{aligned}$$



$$\int_0^1 \int_{x^2}^x 6 \, dy \, dx$$

$f_X(x)$

$$\int_0^1 \int_y^{\sqrt{y}} 6 \, dx \, dy$$

$f_Y(y)$

b) X, Y son indep. $\Leftrightarrow f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$

$$6 \neq [6x(1-x)] [6(\sqrt{y}-y)]$$

$\therefore X, Y$ no son independientes.

$$6. f_{X,Y}(X,Y) = \begin{cases} 30XY^2 & \text{si } 0 \leq X \leq 1, X-1 \leq Y \leq 1-X \\ 0 & \text{en o.c.} \end{cases}$$

$$f_X(X) = \int_{X-1}^{1-X} 30XY^2 dY$$

$$= 30X \cdot \frac{Y^3}{3} \Big|_{X-1}^{1-X}$$

$$= 10X \cdot [(1-X)^3 - (X-1)^3]$$

$$= 10X \cdot [(1-X)^3 + (1-X)^3]$$

$$= 10X [2(1-X)^3]$$

$$\boxed{f_X(X) = 20X(1-X)^3}$$

$$f_Y(Y) = \int_0^{Y+1} 30XY^2 dX$$

$$= 30Y^2 \cdot \frac{X^2}{2} \Big|_0^{Y+1}$$

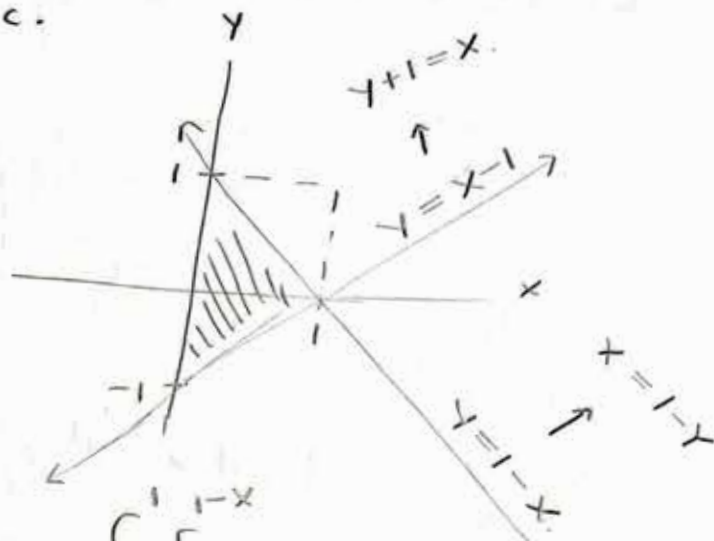
$$\boxed{f_Y(Y) = 15Y^2(Y+1)^2 \text{ si } -1 < Y < 0}$$

$$f_Y(Y) = \int_0^{1-Y} 30XY^2 dX$$

$$= 30Y^2 \cdot \frac{X^2}{2} \Big|_0^{1-Y}$$

$$\boxed{f_Y(Y) = 15Y^2(1-Y)^2 \text{ si } 0 < Y < 1}$$

$$f_Y(Y) = \begin{cases} 15Y^2(Y+1)^2 & \text{si } -1 < Y < 0 \\ 15Y^2(1-Y)^2 & \text{si } 0 < Y < 1 \\ 0 & \text{en o.c.} \end{cases}$$



$$\int_0^1 \int_{X-1}^{1-X} 30XY^2 dY dX$$

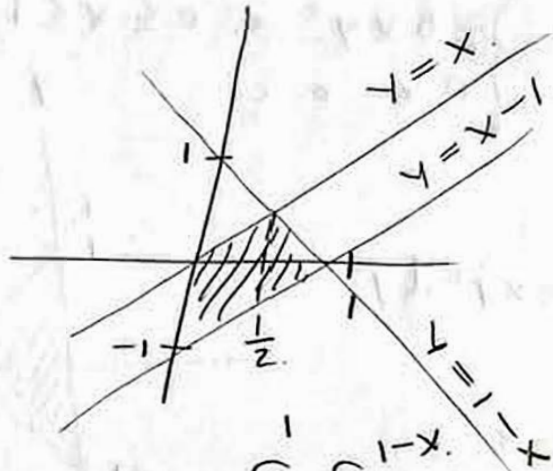
$$\int_{-1}^0 \int_0^{Y+1} 30XY^2 dX dY$$

$$\int_0^1 \int_0^{1-Y} 30XY^2 dX dY$$

$$f_Y(Y) \text{ si } 0 < Y < 1$$

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

b) $P(X > Y)$



$$= \int_0^{\frac{1}{2}} \int_{x-1}^x 30xy^2 dy dx + \int_{\frac{1}{2}}^1 \int_{x-1}^{1-x} 30xy^2 dy dx$$

$$= \int_0^{\frac{1}{2}} 30x \cdot \frac{y^3}{3} \Big|_{x-1}^x dx + \int_{\frac{1}{2}}^1 30x \cdot \frac{y^3}{3} \Big|_{x-1}^{1-x} dx$$

$$= \int_0^{\frac{1}{2}} 10x (x^3 - (x-1)^3) dx + \int_{\frac{1}{2}}^1 10x [(1-x)^3 - (x-1)^3] dx$$

$$= \int_0^{\frac{1}{2}} 10x (x^3 - (x^3 - 3x^2 + 3x - 1)) dx + \int_{\frac{1}{2}}^1 10x [(1-x)^3 + (1-x)^3] dx$$

$$= \int_0^{\frac{1}{2}} 10x (x^3 - x^3 + 3x^2 - 3x + 1) dx + \int_{\frac{1}{2}}^1 10x [2(1-x)^3] dx$$

$$= \int_0^{\frac{1}{2}} 10x (3x^2 - 3x + 1) dx + \int_{\frac{1}{2}}^1 20x (1-x)^3 dx$$

$$= \int_0^{\frac{1}{2}} (30x^3 - 30x^2 + 10x) dx + \int_{\frac{1}{2}}^1 20x (1^3 - 3 \cdot 1^2 \cdot x + 3 \cdot 1 \cdot x^2 - x^3) dx$$

$$= 30 \cdot \frac{x^4}{4} - 30 \cdot \frac{x^3}{3} + 10 \cdot \frac{x^2}{2} \Big|_0^{\frac{1}{2}} + \int_{\frac{1}{2}}^1 20x (1 - 3x + 3x^2 - x^3) dx$$

$$= \frac{15}{32} + \int_{\frac{1}{2}}^1 20x - 60x^2 + 60x^3 - 20x^4 dx$$

$$= \frac{15}{32} + 20 \cdot \frac{x^2}{2} - 60 \cdot \frac{x^3}{3} + 60 \cdot \frac{x^4}{4} - 20 \cdot \frac{x^5}{5} \Big|_{\frac{1}{2}}^1$$

$$= \frac{15}{32} + \left(1 - \frac{13}{16}\right)$$

$$= \frac{15}{32} + \frac{3}{16}$$

$$\boxed{= \frac{21}{32}}$$

$$c) F_{Y|X}(y|x) = \int_{-\infty}^y \frac{f_{X,Y}(x,y)}{f_X(x)} dt$$

$$= \int_{x-1}^y \frac{30xt^2}{20x(1-x)^3} dt$$

$$= \int_{x-1}^y \frac{3t^2}{2(1-x)^3} dt$$

$$= \frac{1}{2(1-x)^3} \cdot 3 \cdot \frac{t^3}{3} \Big|_{(x-1)}^y$$

$$= \frac{1}{2(1-x)^3} \cdot (y^3 - (x-1)^3)$$

$$\boxed{= \frac{y^3 - (x-1)^3}{2(1-x)^3}}$$

$$d. P(Y > 0 | X = \frac{3}{4})$$

$$= \int_0^{1 - \frac{3}{4} = \frac{1}{4}} 30 \cdot \frac{3}{4} y^2 dy$$

$$20 \cdot \frac{3}{4} \left(1 - \frac{3}{4}\right)^3$$

$$= \frac{15}{128}$$

$$\frac{15}{64}$$

$$\boxed{= \frac{1}{2_4}}$$

$$e) X, Y \text{ son indep} \Leftrightarrow f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$$

$$30xy^2 \neq 20x(1-x)^3 \cdot 15y^2(y+1)^2 \cdot 15y^2(1-y)^2$$

$$\therefore X, Y \text{ no son independientes,}$$

$$7. f_{X,Y}(x,y) = \begin{cases} \frac{1}{y} \cdot e^{-\frac{x}{y}-y} & \text{si } x > 0, y > 0 \\ 0 & \text{en o.c.} \end{cases}$$

a) $f_X(x) = \text{no sale}$

$$f_Y(y) = \int_0^{\infty} \frac{1}{y} \cdot e^{-\frac{x}{y}-y} dx$$

$$= \frac{1}{y} \cdot e^{-y} \int_0^{\infty} e^{-\frac{x}{y}} dx$$

$$= \frac{1}{y} \cdot e^{-y} \cdot \left[\frac{e^{-\frac{x}{y}}}{-\frac{1}{y}} \right]_0^{\infty}$$

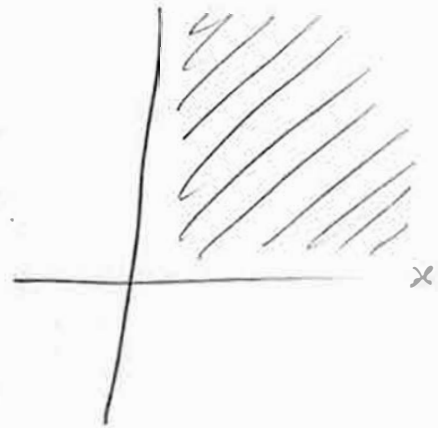
$$= \frac{1}{y} \cdot e^{-y} \cdot \left[-y e^{-\frac{x}{y}} \right]_0^{\infty}$$

$$= \frac{1}{y} \cdot e^{-y} (0 - (-y))$$

$$= \frac{1}{y} \cdot y \cdot e^{-y}$$

$$= \frac{y}{y e^y}$$

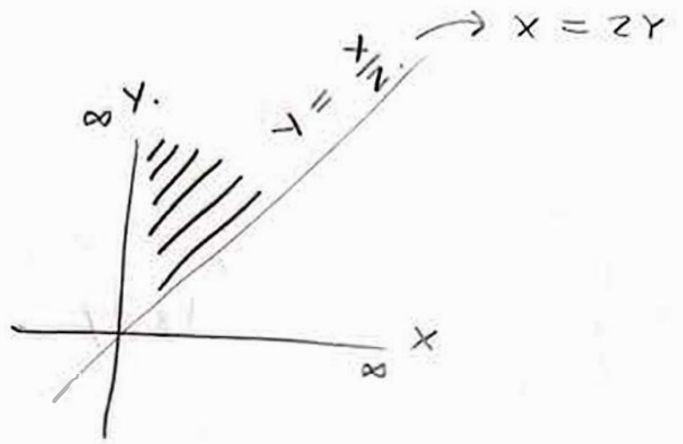
$$\left[= \frac{1}{e^y} \right]$$



Práctica 2

Parte 2

$$\begin{aligned}
 b) P(X < 2Y) \\
 &= P\left(\frac{X}{2} < Y\right) \\
 &= P\left(Y > \frac{X}{2}\right)
 \end{aligned}$$



$$= \int_0^{\infty} \int_0^{2y} \frac{1}{y} \cdot e^{-\frac{x}{y}} - y \, dx \, dy$$

$$= \int_0^{\infty} \frac{1}{y} \cdot e^{-y} \int_0^{2y} e^{-\frac{x}{y}} \, dx \, dy$$

$$= \int_0^{\infty} \frac{1}{y e^y} \cdot \left[\frac{e^{-\frac{x}{y}}}{-\frac{1}{y}} \right]_0^{2y} \, dy$$

$$= \int_0^{\infty} \frac{1}{y e^y} \cdot -y e^{-\frac{x}{y}} \Big|_0^{2y} \, dy$$

$$= \int_0^{\infty} \frac{1}{y e^y} (-y e^{-\frac{2y}{y}} - -y) \, dy$$

$$= \int_0^{\infty} \frac{1}{y e^y} (-y e^{-2} + y) \, dy$$

$$= \int_0^{\infty} \frac{-y e^{-2}}{y e^y} + \frac{y}{y e^y} \, dy$$

$$= \int_0^{\infty} \frac{-e^{-2} + 1}{e^y} \, dy$$

$$= -e^{-2} + 1 \int_0^{\infty} e^{-y} \, dy$$

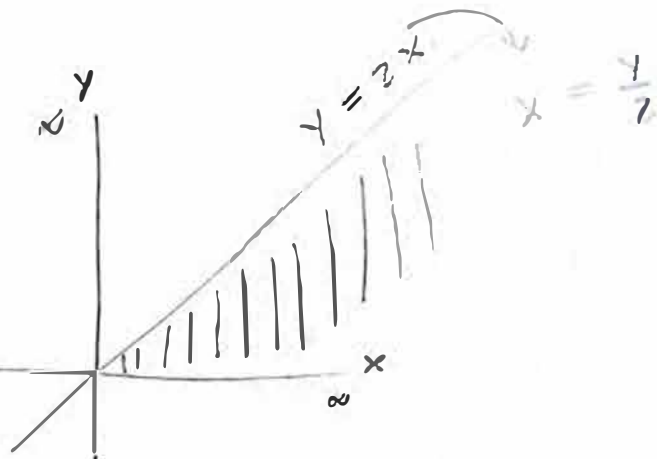
$$= -e^{-2} + 1 - e^{-y} \Big|_0^{\infty}$$

$$= -e^{-2} + 1 (0 - -1) = -e^{-2} + 1 \approx \boxed{0,8647,}$$

(2)

$$c) P(Y < 2X)$$

$$= \int_0^{\infty} \int_{\frac{y}{2}}^{\infty} \frac{1}{y} \cdot e^{-\frac{x}{y}} - y \, dx \, dy$$



$$= \int_0^{\infty} \frac{1}{y} \cdot e^{-y} \int_{\frac{y}{2}}^{\infty} e^{-\frac{x}{y}} \, dx \, dy$$

$$= \int_0^{\infty} \frac{1}{y e^y} \cdot \left. \frac{e^{-\frac{x}{y}}}{-\frac{1}{y}} \right|_{\frac{y}{2}}^{\infty} dy$$

$$= \int_0^{\infty} \frac{1}{y e^y} \cdot -y e^{-\frac{x}{y}} \bigg|_{\frac{y}{2}}^{\infty} dy$$

$$= \int_0^{\infty} \frac{1}{y e^y} (0 + y \cdot e^{-\frac{\frac{y}{2}}{y}}) dy$$

$$= \int_0^{\infty} \frac{1}{y e^y} (0 + y e^{-\frac{1}{2}}) dy$$

$$= \int_0^{\infty} \frac{y e^{-\frac{1}{2}}}{y e^y} dy$$

$$= e^{-\frac{1}{2}} \int_0^{\infty} e^{-y} dy$$

$$= e^{-\frac{1}{2}} \cdot -e^{-y} \bigg|_0^{\infty}$$

$$= e^{-\frac{1}{2}} (0 - -1)$$

$$= e^{-\frac{1}{2}} (\approx 0,6065)$$

8.

$$f_{x,y}(x,y) = \frac{d^2}{dy dx} \left\{ \frac{1}{2} (x^3 y + x y^3) \right\}$$

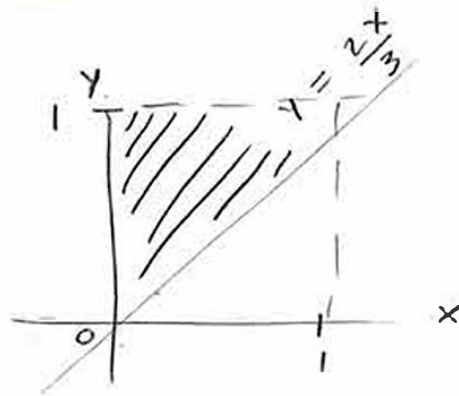
$$= \frac{d}{dy} \left\{ \frac{1}{2} (3x^2 y + y^3) \right\}$$

$$= \frac{1}{2} (3x^2 + 3y^2)$$

$$= \frac{3}{2} (x^2 + y^2)$$

$$f_{x,y}(x,y) = \begin{cases} \frac{3}{2} (x^2 + y^2) & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{on o.c.} \end{cases}$$

$$\begin{aligned}
 b) P(2X < 3Y) \\
 &= P\left(\frac{2X}{3} < Y\right) \\
 &= P\left(Y > \frac{2X}{3}\right)
 \end{aligned}$$



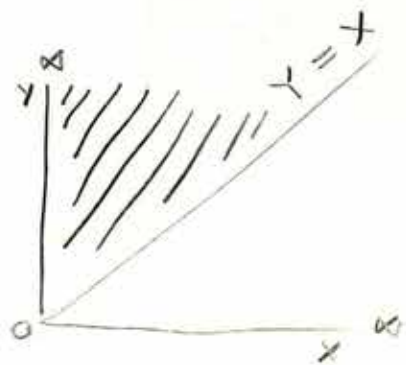
$$\begin{aligned}
 &= \int_0^1 \int_{\frac{2x}{3}}^1 \left(\frac{3}{2} x^2 + \frac{3}{2} y^2 \right) dy dx \\
 &= \int_0^1 \left(\frac{3}{2} x^2 \cdot y + \frac{3}{2} \cdot \frac{y^3}{3} \right) \bigg|_{\frac{2x}{3}}^1 dx \\
 &= \int_0^1 \left(\frac{3x^2}{2} + \frac{3}{6} - \left(\frac{3x^2}{2} \cdot \left(\frac{2x}{3} \right) + \frac{3}{2} \cdot \frac{\left(\frac{2x}{3} \right)^3}{3} \right) \right) dx \\
 &= \int_0^1 \left(\frac{3x^2}{2} + \frac{1}{2} - \left(\frac{6x^3}{6} + \frac{3}{2} \cdot \frac{\frac{2x^3}{27}}{1} \right) \right) dx \\
 &= \int_0^1 \left(\frac{3x^2}{2} + \frac{1}{2} - \left(x^3 + \frac{3}{2} \cdot \frac{8x^3}{81} \right) \right) dx \\
 &= \int_0^1 \left(\frac{3x^2}{2} + \frac{1}{2} - x^3 - \frac{4}{27} x^3 \right) dx \\
 &= \left(\frac{3}{2} \cdot \frac{x^3}{3} + \frac{1}{2} \cdot x - \frac{x^4}{4} - \frac{4}{27} \cdot \frac{x^4}{4} \right) \bigg|_0^1
 \end{aligned}$$

$$\boxed{\approx 0,7129}$$

$$9. f_X(x) = \lambda_1 \cdot e^{-\lambda_1 \cdot x}$$

$$f_Y(y) = \lambda_2 \cdot e^{-\lambda_2 \cdot y}$$

$$f_{X,Y}(x,y) = \begin{cases} \lambda_1 \cdot \lambda_2 \cdot e^{-\lambda_1 \cdot x} \cdot e^{-\lambda_2 \cdot y} & \text{si } 0 \leq x \leq \infty \text{ y } 0 \leq y \leq \infty \\ 0 & \text{en o.c.} \end{cases}$$



$$P(X < Y)$$

$$= P(Y > X)$$

$$= \int_0^{\infty} \int_x^{\infty} \lambda_1 \cdot \lambda_2 \cdot e^{-\lambda_1 \cdot x} \cdot e^{-\lambda_2 \cdot y} dy dx$$

$$= \lambda_1 \cdot \lambda_2 \int_0^{\infty} e^{-\lambda_1 \cdot x} \int_x^{\infty} e^{-\lambda_2 \cdot y} dy dx$$

$$= \lambda_1 \cdot \lambda_2 \int_0^{\infty} e^{-\lambda_1 \cdot x} \cdot \left[\frac{e^{-\lambda_2 \cdot y}}{-\lambda_2} \right]_x^{\infty} dx$$

$$= -\frac{\lambda_1 \cdot \lambda_2}{\lambda_2} \int_0^{\infty} e^{-\lambda_1 \cdot x} (0 - e^{-\lambda_2 \cdot x}) dx$$

$$= -\lambda_1 \int_0^{\infty} e^{-\lambda_1 \cdot x} \cdot -e^{-\lambda_2 \cdot x} dx$$

$$= \lambda_1 \cdot \int_0^{\infty} e^{-\lambda_1 \cdot x - \lambda_2 \cdot x} dx$$

$$= \lambda_1 \cdot \int_0^{\infty} e^{-(\lambda_1 + \lambda_2) \cdot x} dx$$

$$= \lambda_1 \cdot \left[\frac{e^{-(\lambda_1 + \lambda_2) \cdot x}}{-(\lambda_1 + \lambda_2)} \right]_0^{\infty} = \lambda_1 \cdot \left(0 - -\frac{1}{\lambda_1 + \lambda_2} \right)$$

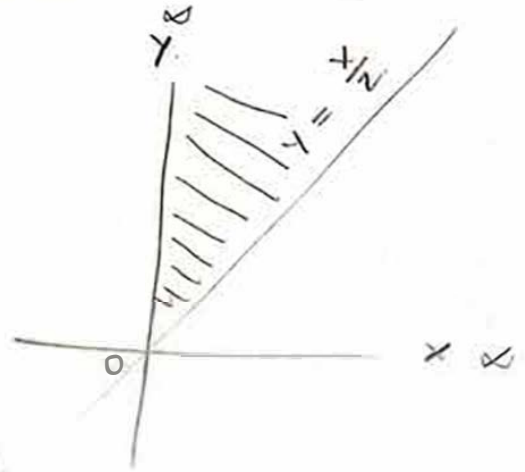
$$= \left[\frac{\lambda_1}{\lambda_1 + \lambda_2} \right]$$

(5)

$$10. f_X(x) = \lambda \cdot e^{-\lambda x}.$$

$$f_Y(y) = \lambda \cdot e^{-\lambda y}.$$

$$f_{X,Y}(x,y) = \begin{cases} \lambda^2 \cdot e^{-\lambda x} \cdot e^{-\lambda y} & \text{si } x > 0, y > 0 \\ 0 & \text{en o. c.} \end{cases}$$



$$a) P(X < 2Y)$$

$$= P\left(\frac{X}{2} < Y\right)$$

$$= P\left(Y > \frac{X}{2}\right)$$

$$= \int_0^{\infty} \int_{\frac{x}{2}}^{\infty} \lambda^2 \cdot e^{-\lambda x} \cdot e^{-\lambda y} dy dx$$

$$= \lambda^2 \int_0^{\infty} e^{-\lambda x} \int_{\frac{x}{2}}^{\infty} e^{-\lambda y} dy dx$$

$$= \lambda^2 \int_0^{\infty} e^{-\lambda x} \left[\frac{e^{-\lambda y}}{-\lambda} \right]_{\frac{x}{2}}^{\infty} dx$$

$$= -\lambda \int_0^{\infty} e^{-\lambda x} (0 - e^{-\frac{\lambda x}{2}}) dx$$

$$= \lambda \int_0^{\infty} e^{-\lambda x - \frac{\lambda x}{2}} dx$$

$$= \lambda \int_0^{\infty} e^{\frac{-2\lambda x - \lambda x}{2}} dx$$

$$= \lambda \int_0^{\infty} e^{\frac{-3\lambda x}{2}} dx$$

$$= \lambda \cdot \left[\frac{e^{\frac{-3\lambda x}{2}}}{-\frac{3\lambda}{2}} \right]_0^{\infty}$$

$$= \lambda \left(0 - -\frac{1}{\frac{3\lambda}{2}} \right) = \frac{\lambda}{\frac{3\lambda}{2}} = \frac{2}{3} = \boxed{\frac{2}{3}} \quad (6)$$

$$\begin{aligned}
 b) P(X+Y \leq 1, X-Y \leq 1) \\
 = P(Y \leq 1-X, X-1 \leq Y) \\
 = P(Y \leq 1-X, Y \geq X-1)
 \end{aligned}$$

$$= \int_0^1 \int_0^{1-x} \lambda^2 \cdot e^{-\lambda x} \cdot e^{-\lambda y} dy dx$$

$$= \lambda^2 \int_0^1 e^{-\lambda x} \int_0^{1-x} e^{-\lambda y} dy dx$$

$$= -\lambda^2 \int_0^1 e^{-\lambda x} \left[\frac{e^{-\lambda y}}{-\lambda} \right]_0^{1-x} dx$$

$$= -\lambda \int_0^1 e^{-\lambda x} (e^{-\lambda(1-x)} - 1) dx$$

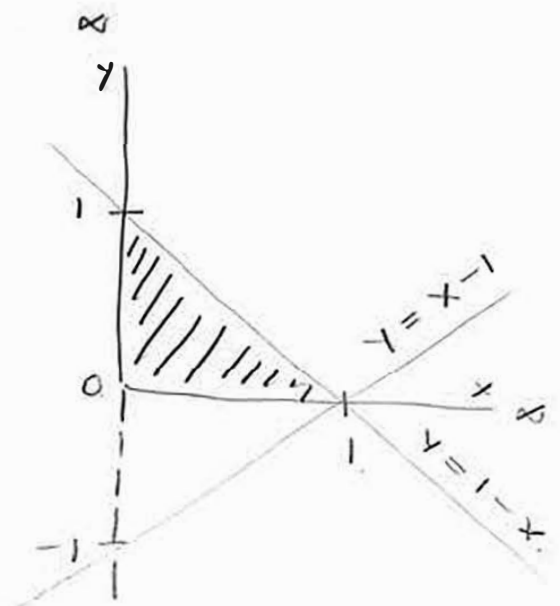
$$= -\lambda \int_0^1 e^{-\lambda x} (e^{-\lambda + \lambda x} - 1) dx$$

$$= -\lambda \int_0^1 e^{-\lambda x - \lambda + \lambda x} - e^{-\lambda x} dx$$

$$= -\lambda \cdot x \cdot e^{-\lambda} + \frac{e^{-\lambda x}}{\lambda} \Big|_0^1$$

$$= -\lambda \left[\left(e^{-\lambda} + \frac{e^{-\lambda}}{\lambda} \right) - \left(0 + \frac{1}{\lambda} \right) \right]$$

$$\begin{aligned}
 &= -\lambda \cdot \left[e^{-\lambda} + \frac{e^{-\lambda}}{\lambda} - \frac{1}{\lambda} \right] = -\lambda e^{-\lambda} - \frac{\lambda e^{-\lambda}}{\lambda} + \frac{\lambda}{\lambda} \\
 &= \boxed{-\lambda e^{-\lambda} - e^{-\lambda} + 1} \quad (7)
 \end{aligned}$$



$$11. f_{X|Y}(x|y) = \begin{cases} \gamma e^{-xy} & \text{si } x \geq 0. \\ 0 & \text{en o. c.} \end{cases}$$

$$f_Y(y) = \begin{cases} e^{-y} & \text{si } y > 0. \\ 0 & \text{en o. c.} \end{cases}$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

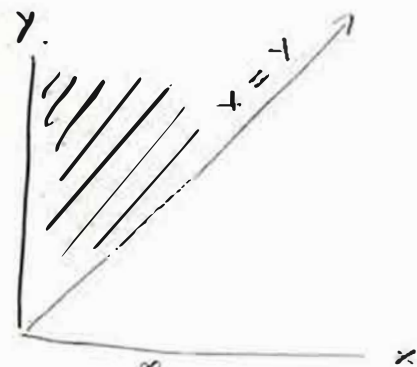
$$\Rightarrow \gamma e^{-xy} = \frac{f_{X,Y}(x,y)}{e^{-y}}$$

$$\Rightarrow \gamma e^{-xy-y} = f_{X,Y}(x,y)$$

$$\Rightarrow \gamma e^{-y(x+1)} = f_{X,Y}(x,y)$$

$$\therefore f_{X,Y}(x,y) = \begin{cases} \gamma e^{-y(x+1)} & \text{si } x > 0, y > 0. \\ 0 & \text{en o. c.} \end{cases}$$

$$12. f_{X,Y}(x,y) = \begin{cases} e^{-y}, & 0 \leq x < y < \infty \\ 0 & \text{on o.c.} \end{cases}$$



$$P(X \leq 1 | Y \geq 2) = \frac{\int_2^{\infty} \int_0^1 e^{-y} dx dy}{\int_2^{\infty} y e^{-y} dy}$$

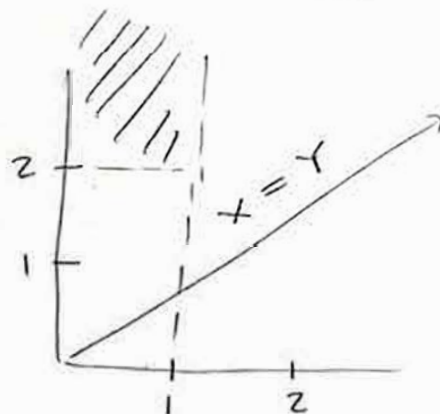
$$f_Y(y) = \int_0^y e^{-y} dx dy = \int_0^y 1 \cdot e^{-y} dx dy = \int_0^y e^{-y} dx dy$$

$$= \int_2^{\infty} x \cdot e^{-y} \Big|_0^1 dy = \int_2^{\infty} y \cdot e^{-y} dy$$

$u = y \quad dv = e^{-y} dy$
 $du = dy \quad v = -e^{-y}$

$$= \int_0^y e^{-y} dx = x \cdot e^{-y} \Big|_0^y = y e^{-y}$$

$$= \int_2^{\infty} e^{-y} dy = -y e^{-y} - \int -e^{-y} dy$$



$$= -e^{-y} \Big|_2^{\infty} = -y e^{-y} + -e^{-y} \Big|_2^{\infty}$$

$$= \frac{0 - (-e^{-2})}{(0 + 0 - (-2e^{-2} - e^{-2}))} = \frac{e^{-2}}{2e^{-2} + e^{-2}} = \frac{e^{-2}}{3e^{-2}} = \frac{1}{3}$$

$$= \frac{e^{-2}}{2e^{-2} + e^{-2}} = \frac{e^{-2}}{3e^{-2}} = \frac{1}{3}$$

(9)

$$13. f_{X,Y}(x,y) = \begin{cases} e^{-x} & \text{si } 0 \leq y \leq x \\ 0 & \text{m.o.c.} \end{cases}$$

$$W = X - Y$$

$$a) E(W)$$

$$= E(X - Y)$$

$$= E_X(X) - E_Y(Y)$$

$$f_X(X) = \int_0^x e^{-x} dy$$

$$= y \cdot e^{-x} \Big|_0^x$$

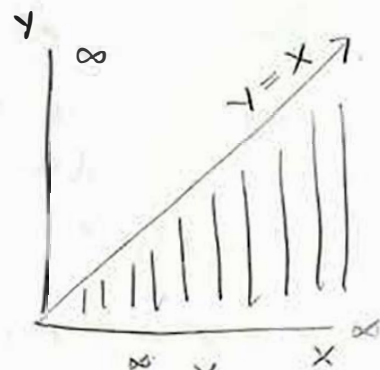
$$= \boxed{x e^{-x}}$$

$$f_Y(Y) = \int_Y^\infty e^{-x} dx$$

$$= -e^{-x} \Big|_Y^\infty$$

$$= 0 - (-e^{-Y})$$

$$= \boxed{e^{-Y}}$$



$$\int_0^\infty \int_0^x e^{-x} dy dx$$

$f_X(X)$

$$\int_0^\infty \int_Y^\infty e^{-x} dx dy$$

$f_Y(Y)$

Práctica 2

Parte 3

$$E_X(X) = \int_0^{\infty} x \cdot x e^{-x} dx.$$

$$= \int_0^{\infty} x^2 \cdot e^{-x} dx.$$

$$u = x^2 \quad dv = e^{-x} dx$$

$$du = 2x dx \quad v = -e^{-x}$$

$$= -x^2 e^{-x} - \int -2x e^{-x} dx$$

$$= -x^2 e^{-x} + 2 \int x e^{-x} dx$$

$$u = x \quad dv = e^{-x}$$

$$du = dx \quad v = -e^{-x}$$

$$= -x^2 e^{-x} + 2 \left(-x e^{-x} - \int -e^{-x} dx \right)$$

$$= -x^2 e^{-x} + 2 \left(-x e^{-x} + -e^{-x} \right) \Big|_0^{\infty}$$

$$= -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} \Big|_0^{\infty}$$

$$= 0 - (-2 \cdot 1)$$

$$\boxed{= 2,}$$

$$E_Y(Y) = \int_0^{\infty} y \cdot e^{-y} dy$$

$$u = y \quad dv = e^{-y}$$

$$du = dy \quad v = -e^{-y}$$

$$= -y e^{-y} - \int -e^{-y} dy$$

$$= -y e^{-y} + -e^{-y} \Big|_0^{\infty}$$

$$= 0 - (-1)$$

$$\boxed{= 1,}$$

$$\therefore E(W)$$

$$= E(X - Y)$$

$$= E(X) - E(Y)$$

$$= \boxed{\frac{2-1}{1,}}$$

$$E(X^2) = \int_0^{\infty} x^2 \cdot x e^{-x} dx$$

$$= \int_0^{\infty} x^3 \cdot e^{-x} dx$$

$$u = x^3 \quad dv = e^{-x} dx$$

$$du = 3x^2 dx \quad v = -e^{-x}$$

$$= -x^3 e^{-x} - \int -3x^2 e^{-x} dx$$

$$= -x^3 e^{-x} + 3 \int x^2 e^{-x} dx$$

$$u = x^2 \quad dv = e^{-x} dx$$

$$du = 2x dx \quad v = -e^{-x}$$

$$= -x^3 e^{-x} + 3 \left(-x^2 e^{-x} - \int -2x e^{-x} dx \right)$$

$$= -x^3 e^{-x} + 3 \left(-x^2 e^{-x} + 2 \int x e^{-x} dx \right)$$

$$u = x$$

$$dv = e^{-x}$$

$$du = dx$$

$$v = -e^{-x}$$

$$= -x^3 e^{-x} + 3 \left(-x^2 e^{-x} + 2 \left(-x e^{-x} - \int -e^{-x} dx \right) \right)$$

$$= -x^3 e^{-x} + 3 \left(-x^2 e^{-x} + 2 \left(-x e^{-x} + -e^{-x} \right) \right) \Big|_0^{\infty}$$

$$= -x^3 e^{-x} + 3 \left(-x^2 e^{-x} - 2x e^{-x} - 2e^{-x} \right) \Big|_0^{\infty}$$

$$= -x^3 e^{-x} - 3x^2 e^{-x} - 6x e^{-x} - 6e^{-x} \Big|_0^{\infty}$$

$$= 0 - (0 - 6 \cdot 1)$$

$$\boxed{= 6,,}$$

$$E(Y^2) = \int_0^{\infty} y^2 \cdot e^{-y} dy$$

$$u = y^2 \quad dv = e^{-y} dy$$

$$du = 2y dy \quad v = -e^{-y}$$

$$= -y^2 e^{-y} - \int -2y e^{-y} dy$$

$$= -y^2 e^{-y} + 2 \int y e^{-y} dy$$

$$u = y \quad dv = e^{-y} dy$$

$$du = dy \quad v = -e^{-y}$$

$$= -y^2 e^{-y} + 2(-y e^{-y} - \int -e^{-y} dy)$$

$$= -y^2 e^{-y} + 2(-y e^{-y} + e^{-y}) \Big|_0^{\infty}$$

$$= -y^2 e^{-y} - 2y e^{-y} - 2e^{-y} \Big|_0^{\infty}$$

$$= 0 - (-2 \cdot 1)$$

$$\boxed{= 2}$$

$$E_{XY}(XY) = \int_0^{\infty} \int_0^x x \cdot y \cdot e^{-x} dy dx$$

$$= \int_0^{\infty} x \cdot e^{-x} \int_0^x y dy dx$$

$$= \int_0^{\infty} x \cdot e^{-x} \cdot \frac{y^2}{2} \Big|_0^x dx$$

$$= \int_0^{\infty} x \cdot e^{-x} \cdot \frac{x^2}{2} dx$$

$$= \frac{1}{2} \int_0^{\infty} x^3 \cdot e^{-x} dx$$

$$u = x^3 \quad dv = e^{-x} dx$$

$$du = 3x^2 dx \quad v = -e^{-x}$$

$$= \frac{1}{2} \left(-x^3 e^{-x} - \int -3x^2 e^{-x} dx \right)$$

$$= \frac{1}{2} \left(-x^3 e^{-x} + 3 \int x^2 \cdot e^{-x} dx \right)$$

$$u = x^2 \quad dv = e^{-x} dx$$

$$du = 2x dx \quad v = -e^{-x}$$

$$\begin{aligned}
&= \frac{1}{2} \left(-x^3 e^{-x} + 3 \left(-x^2 e^{-x} - \int -2x e^{-x} dx \right) \right) \\
&= \frac{1}{2} \left(-x^3 e^{-x} + 3 \left(-x^2 e^{-x} + 2 \int x e^{-x} dx \right) \right) \\
&\quad \begin{matrix} u=x & dv=e^{-x} \\ du=dx & v=-e^{-x} \end{matrix} \\
&= \frac{1}{2} \left(-x^3 e^{-x} + 3 \left(-x^2 e^{-x} + 2 \left(-x e^{-x} - \int -e^{-x} dx \right) \right) \right) \\
&= \frac{1}{2} \left(-x^3 e^{-x} + 3 \left(-x^2 e^{-x} + 2 \left(-x e^{-x} + e^{-x} \right) \right) \right) \Big|_0^\infty \\
&= \frac{1}{2} \left(-x^3 e^{-x} + 3 \left(-x^2 e^{-x} - 2x e^{-x} - 2e^{-x} \right) \right) \Big|_0^\infty \\
&= \frac{1}{2} \left(-x^3 e^{-x} - 3x^2 e^{-x} - 6x e^{-x} - 6e^{-x} \right) \Big|_0^\infty \\
&= \frac{1}{2} (0 - (-6 \cdot 1)) \\
&= \frac{6}{2} \\
&= \boxed{3}
\end{aligned}$$



$$\begin{aligned}
\text{Var}(W) &= \text{Var}(X - Y) \\
&= \text{Var}(X) + \text{Var}(Y) - 2 \text{Cov}(X, Y) \\
&= [6 - (2)^2] + [2 - (1)^2] - [2 \cdot (3 - 2 \cdot 1)] \\
&= [6 - 4] + [2 - 1] - [2 \cdot 1] \\
&= 2 + 1 - 2 \\
&= \boxed{1}
\end{aligned}$$

Nota :

$$\text{Var}(X - Y) = (1)^2 \cdot \text{Var}(X) + (-1)^2 \text{Var}(Y) + 2(1)(-1) \text{Cov}(X, Y)$$

$$b) P(X - Y \geq 4)$$

$$= P(X \geq 4 + Y)$$

$$= \int_0^{\infty} \int_{y+4}^{\infty} e^{-x} dx dy$$

$$= \int_0^{\infty} -e^{-x} \Big|_{y+4}^{\infty} dy$$

$$= \int_0^{\infty} (0 - -e^{-(y+4)}) dy$$

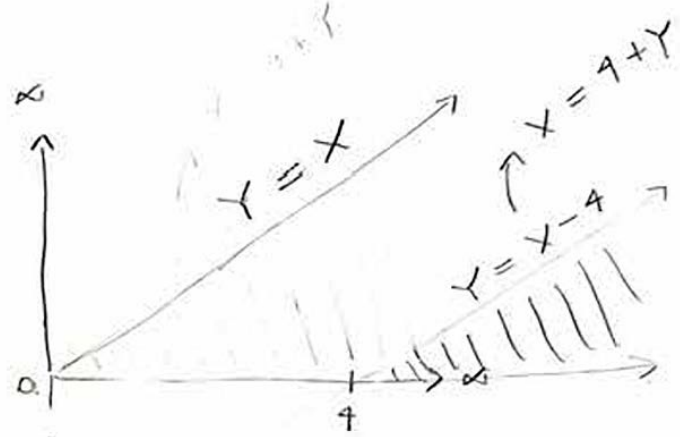
$$= \int_0^{\infty} e^{-y} \cdot e^{-4} dy$$

$$= e^{-4} \int_0^{\infty} e^{-y} dy$$

$$= e^{-4} \cdot -e^{-y} \Big|_0^{\infty}$$

$$= e^{-4} (0 - -1)$$

$$\boxed{= e^{-4}}$$



$$14. f_{X,Y}(X,Y) = \begin{cases} e^{-x-y} & \text{si } x > 0, y > 0 \\ 0 & \text{en o.c.} \end{cases}$$

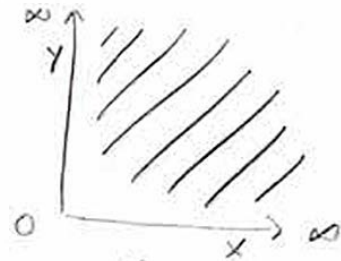
$$a) f_X(x) = \int_0^{\infty} e^{-x} \cdot e^{-y} dy$$

$$= e^{-x} \int_0^{\infty} e^{-y} dy$$

$$= e^{-x} \cdot -e^{-y} \Big|_0^{\infty}$$

$$= e^{-x} (0 - -1)$$

$$\boxed{f_X(x) = e^{-x}} \quad \text{si } x > 0.$$



$$\int_0^{\infty} \int_0^{\infty} e^{-x-y} dx dy$$

$$\int_0^{\infty} \int_0^{\infty} e^{-x-y} dy dx$$

$$f_Y(y) = \int_0^{\infty} e^{-x} \cdot e^{-y} dx$$

$$= e^{-y} \int_0^{\infty} e^{-x} dx$$

$$= e^{-y} \cdot -e^{-x} \Big|_0^{\infty}$$

$$= e^{-y} (0 - -1)$$

$$\boxed{f_Y(y) = e^{-y}} \quad y > 0.$$

$$b) P(0 < X < 1, 1 \leq Y < 2)$$

$$= \int_0^1 \int_1^2 e^{-x-y} dy dx$$

$$= \int_0^1 e^{-x} \int_1^2 e^{-y} dy dx$$

$$= \int_0^1 e^{-x} \cdot -e^{-y} \Big|_1^2 dx$$

$$= \int_0^1 e^{-x} (-e^{-2} - -e^{-1}) dx$$

$$= (-e^{-2} + e^{-1}) \int_0^1 e^{-x} dx$$

$$= (-e^{-2} + e^{-1}) \cdot -e^{-x} \Big|_0^1$$

$$= (-e^{-2} + e^{-1}) (-e^{-1} - -1)$$

$$\approx 0,1469,$$

$$c) F_{Y|X}(y|x=3) = \int_{-\infty}^y \frac{f_{X,Y}(x,y)}{f_X(x)} dt$$

$$= \int_0^y \frac{\cancel{e^{-3}} \cdot e^{-t}}{\cancel{e^{-3}}} dt$$

$$= \int_0^y e^{-t} dt$$

$$= -e^{-t} \Big|_0^y$$

$$= -e^{-y} - (-1)$$

$$\boxed{= -e^{-y} + 1}$$

$$d) E(X^2 Y)$$

$$= \int_0^{\infty} \int_0^{\infty} x^2 y \cdot e^{-x} \cdot e^{-y} dx dy$$

$$= \int_0^{\infty} y \cdot e^{-y} \int_0^{\infty} x^2 \cdot e^{-x} dx dy$$

$u = x^2 \quad dv = e^{-x} dx$
 $du = 2x dx \quad v = -e^{-x}$

$$= \int_0^{\infty} y \cdot e^{-y} \cdot (-x^2 e^{-x} - \int -2x e^{-x} dx) dy$$

$$= \int_0^{\infty} y \cdot e^{-y} (-x^2 e^{-x} + 2 \int x e^{-x} dx) dy$$

$u = x \quad dv = e^{-x} dx$
 $du = dx \quad v = -e^{-x}$

$$= \int_0^{\infty} y e^{-y} (-x^2 e^{-x} + 2(-x e^{-x} - \int -e^{-x} dx)) dy$$

$$= \int_0^{\infty} y e^{-y} (-x^2 e^{-x} + 2(-x e^{-x} + -e^{-x})) \Big|_0^{\infty} dy$$

$$= \int_0^{\infty} y e^{-y} (-x^2 e^{-x} - 2x e^{-x} - 2e^{-x}) \Big|_0^{\infty} dy$$

$$= \int_0^{\infty} y e^{-y} (0 - (-2 \cdot 1)) dy$$

$$= 2 \int_0^{\infty} y e^{-y} dy$$

$u = y \quad dv = e^{-y}$
 $du = dy \quad v = -e^{-y}$

$$= 2 (-y e^{-y} - \int -e^{-y} dy)$$

$$= 2 (-y e^{-y} + -e^{-y}) \Big|_0^{\infty}$$

$$= 2 (0 - (-1))$$

$$\boxed{= 2}$$

$$1 e). \text{Cov}(X, Y) = E_{XY}(XY) - E_X(X) \cdot E_Y(Y)$$

$$E_{XY}(XY) = \int_0^{\infty} \int_0^{\infty} xy \cdot e^{-x-y} dx dy$$

$$= \int_0^{\infty} y \cdot e^{-y} \int_0^{\infty} x \cdot e^{-x} dx dy$$

$$\begin{aligned} u &= x & dv &= e^{-x} dx \\ du &= dx & v &= -e^{-x} \end{aligned}$$

$$= \int_0^{\infty} y \cdot e^{-y} \left(-x e^{-x} - \int -e^{-x} dx \right) \Big|_0^{\infty} dy$$

$$= \int_0^{\infty} y \cdot e^{-y} \left(-x e^{-x} + -e^{-x} \right) \Big|_0^{\infty} dy$$

$$= \int_0^{\infty} y \cdot e^{-y} (0 - -1) dy$$

$$= \int_0^{\infty} y \cdot e^{-y} dy$$

$$\begin{aligned} u &= y & dv &= e^{-y} dy \\ du &= dy & v &= -e^{-y} \end{aligned}$$

$$= \left(-y e^{-y} - \int -e^{-y} dy \right) \Big|_0^{\infty}$$

$$= \left(-y e^{-y} + -e^{-y} \right) \Big|_0^{\infty}$$

$$= (0 - (0 - 1))$$

$$\underline{\underline{1}}$$

$$\begin{aligned}
 E_{X,Y}(X) &= \int_0^{\infty} x \cdot e^{-x} dx \\
 &\quad \begin{array}{l} u=x \quad dv=e^{-x} dx \\ du=dx \quad v=-e^{-x} \end{array} \\
 &= (-xe^{-x} - \int -e^{-x} dx) \Big|_0^{\infty} \\
 &= (-xe^{-x} + -e^{-x}) \Big|_0^{\infty} \\
 &= (0 - (0 - 1)) \\
 &\quad \boxed{= 1,,}
 \end{aligned}$$

$$\begin{aligned}
 E_{X,Y}(Y) &= \int_0^{\infty} y \cdot e^{-y} dy \\
 &\quad \begin{array}{l} u=y \quad dv=e^{-y} dy \\ du=dy \quad v=-e^{-y} \end{array} \\
 &= (-ye^{-y} - \int -e^{-y} dy) \Big|_0^{\infty} \\
 &= (-ye^{-y} + -e^{-y}) \Big|_0^{\infty} \\
 &= 0 - (0 - 1) \\
 &\quad \boxed{= 1,,}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Cov}(XY) &= E_{X,Y}(XY) - E_{X,Y}(X)E_{X,Y}(Y) \\
 &= 1 - 1 \cdot 1 \\
 &\quad \boxed{= 0,,}
 \end{aligned}$$

Nota: Si X, Y son independientes, $\text{Cov}(XY) = 0$
 Pero si $\text{Cov}(XY) = 0$, no necesariamente son independientes.

f) ¿Son X, Y incorrelacionadas?

$$\text{Coef de correlación } \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\rho = \frac{0}{\sigma_X \sigma_Y}$$

$$\boxed{\rho = 0}$$

$\therefore X, Y$ no son incorrelacionadas.

$$15. f_{X,Y}(X,Y) = \begin{cases} \frac{9}{2} X^2 Y^2 & -1 \leq Y \leq X \leq 1 \\ 0 & \text{w.o.c.} \end{cases}$$

$$a) f_{Y|X}(Y|X) = \frac{f_{X,Y}(X,Y)}{f_X(X)}$$

$$f_X(X) = \int_{-1}^X \frac{9}{2} X^2 Y^2 dY$$

$$= \frac{9}{2} X^2 \int_{-1}^X Y^2 dY$$

$$= \frac{9}{2} X^2 \cdot \left. \frac{Y^3}{3} \right|_{-1}^X$$

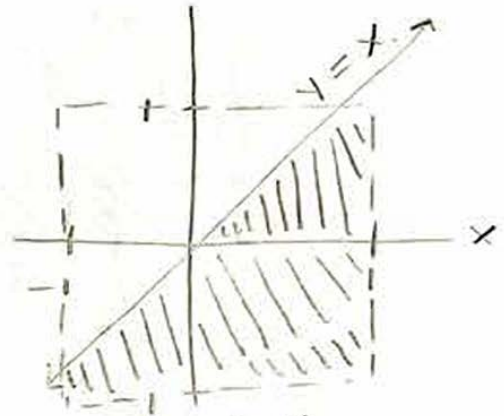
$$= \frac{9}{2} X^2 \left(\frac{X^3}{3} - \frac{(-1)^3}{3} \right)$$

$$= \frac{9}{6} X^2 (X^3 + 1)$$

$$= \left\{ \frac{3}{2} X^5 + \frac{3}{2} X^2 \right\}$$

$$f_{Y|X}(Y|X) = \frac{\frac{9}{2} X^2 Y^2}{\frac{3}{2} X^2 (X^3 + 1)}$$

$$= \left(\frac{3 Y^2}{X^3 + 1} \right)$$



$$\int_{-1}^1 \int_{-1}^X \frac{9}{2} X^2 Y^2 dY dX$$

$$\int_{-1}^1 \underbrace{\int_{-1}^X \frac{9}{2} X^2 Y^2 dY}_{f_X(X)} dX$$

$$\int_{-1}^1 \underbrace{\int_Y^1 \frac{9}{2} X^2 Y^2 dX}_{f_Y(Y)} dY$$

$$b) P(Y \leq 0.1 | X \geq 0.6)$$

$$\int_{0.6}^1 \int_{-1}^{0.1} \frac{9}{2} x^2 y^2 dy dx.$$

$$= \int_{0.6}^1 \left[\frac{3}{2} x^5 + \frac{3}{2} x^2 \right]_{-1}^{0.1} dx$$

$$= \frac{9}{2} \int_{0.6}^1 x^2 \int_{-1}^{0.1} y^2 dy dx$$

$$\frac{9899}{15625}$$

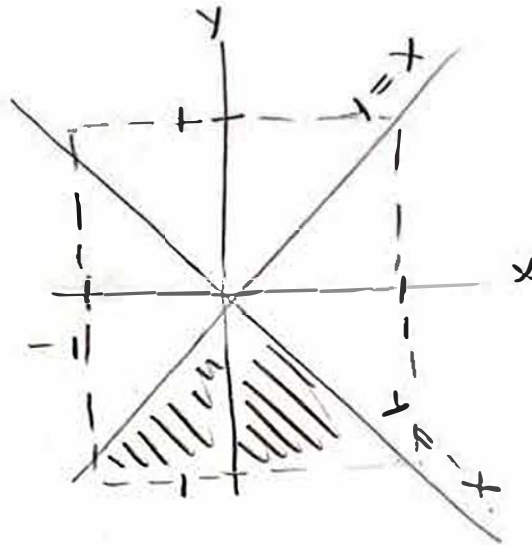
$$= \frac{9}{2} \cdot \frac{1001}{3000} \cdot \frac{98}{375}$$

$$\frac{9899}{15625}$$

$$\boxed{\approx 0,6225,,}$$

$$\begin{aligned} c) P(X+Y \leq 0) \\ = P(Y \leq -X) \end{aligned}$$

$$= \int_{-1}^0 \int_{-Y}^{-Y} \frac{9}{2} x^2 y^2 dx dy.$$



$$= \int_{-1}^0 y^2 \int_{-Y}^{-Y} \frac{9}{2} x^2 dx dy.$$

$$= \frac{9}{2} \int_{-1}^0 y^2 \int_{-Y}^{-Y} x^2 dx dy.$$

$$= \frac{9}{2} \int_{-1}^0 y^2 \left[\frac{x^3}{3} \right]_{-Y}^{-Y} dy.$$

$$= \frac{9}{6} \int_{-1}^0 y^2 (-Y^3 - Y^3) dy.$$

$$= \frac{3}{2} \int_{-1}^0 y^2 (-2Y^3) dy.$$

$$= \frac{3}{2} \int_{-1}^0 -2Y^5 dy.$$

$$= \frac{3}{2} \cdot -2 \cdot \left[\frac{Y^6}{6} \right]_{-1}^0.$$

$$= \frac{3}{2} \left(0 - -\frac{2}{6} \right)$$

$$\boxed{= \frac{1}{2}}$$

$$d) \text{Cov}(X, Y) = E_{XY}(XY) - E_X(X) \cdot E_Y(Y)$$

$$\begin{aligned} f_Y(y) &= \int_{-1}^1 \frac{9}{2} x^2 y^2 dx \\ &= \frac{9}{2} y^2 \cdot \left. \frac{x^3}{3} \right|_{-1}^1 \\ &= \frac{9}{6} y^2 (1 - (-1)^3) \\ &= \left(\frac{3}{2} y^2 - \frac{3}{2} y^5 \right) \end{aligned}$$

$$\begin{aligned} E_X(X) &= \int_{-1}^1 x \left(\frac{3}{2} x^5 + \frac{3}{2} x^2 \right) dx \\ &= \int_{-1}^1 \frac{3}{2} x^6 + \frac{3}{2} x^3 dx \\ &= \frac{3}{2} \cdot \left. \frac{x^7}{7} \right|_{-1}^1 + \frac{3}{2} \cdot \left. \frac{x^4}{4} \right|_{-1}^1 \\ &= \frac{3}{56} - \frac{9}{56} \\ &= \left(\frac{3}{7} \right) \end{aligned}$$

$$\begin{aligned}
 E_Y(Y) &= \int_{-1}^1 y \cdot \left(\frac{3}{2} y^2 - \frac{3}{2} y^5 \right) dy \\
 &= \int_{-1}^1 \frac{3}{2} y^3 - \frac{3}{2} y^6 dy \\
 &= \frac{3}{2} \cdot \frac{y^4}{4} - \frac{3}{2} \cdot \frac{y^7}{7} \Big|_{-1}^1 \\
 &= \frac{9}{56} - \frac{3}{56} \\
 &= \boxed{\frac{-3}{7}}
 \end{aligned}$$

$$\begin{aligned}
 E_{XY}(XY) &= \int_{-1}^1 \int_{-1}^x x \cdot y \cdot \frac{9}{2} x^2 y^2 dy dx \\
 &= \frac{9}{2} \int_{-1}^1 x^3 \int_{-1}^x y^3 dy dx \\
 &= \frac{9}{2} \int_{-1}^1 x^3 \cdot \frac{y^4}{4} \Big|_{-1}^x dx \\
 &= \frac{9}{8} \int_{-1}^1 x^3 \cdot (x^4 - (-1)^4) dx \\
 &= \frac{9}{8} \int_{-1}^1 x^3 (x^4 - 1) dx \\
 &= \frac{9}{8} \int_{-1}^1 x^7 - x^3 dx \\
 &= \frac{9}{8} \cdot 0 \\
 &= \boxed{0}
 \end{aligned}$$

$$\begin{aligned}\therefore \text{Cov}(X, Y) &= E_{XY}(XY) - E_X(X) \cdot E_Y(Y) \\ &= 0 - \frac{3}{7} \cdot -\frac{3}{7} \\ &= \frac{9}{49}\end{aligned}$$

$$e) \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\begin{aligned}E(X^2) &= \int_{-1}^1 x^2 \left(\frac{3}{2} x^5 + \frac{3}{2} x^2 \right) dx \\ &= \int_{-1}^1 \frac{3}{2} x^7 + \frac{3}{2} x^4 dx \\ &= \left[\frac{3}{5} \right]\end{aligned}$$

$$\begin{aligned}\text{Var}(X) &= \frac{3}{5} - \left(\frac{3}{7} \right)^2 \\ &= \frac{102}{245}\end{aligned}$$

$$\sigma_X = \sqrt{\frac{102}{245}}$$

$$V_{\alpha}(Y) = E(Y^2) - [E(Y)]^2$$

$$E(Y^2) = \int_{-1}^1 y^2 \left(\frac{3}{2} y^2 - \frac{3}{2} y^5 \right) dy$$

$$= \int_{-1}^1 \frac{3}{2} y^4 - \frac{3}{2} y^7 dy$$

$$= \left[\frac{3}{5} y^5 - \frac{3}{14} y^8 \right]_{-1}^1$$

$$V_{\alpha}(Y) = \frac{3}{5} - \left(-\frac{3}{7} \right)^2$$

$$= \left[\frac{102}{245} \right]$$

$$\sigma_Y = \sqrt{\frac{102}{245}}$$

$$\therefore \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$= \frac{9}{\sqrt{\frac{102}{245}}} \cdot \frac{1}{\sqrt{\frac{102}{245}}}$$

$$= \frac{15}{34} \approx 0,4411$$